PHYSICAL CHEMISTRY

GUIDED REVISION PLAN_SCORE ADVANCED [SOLUTION]

Topic : Chemical Equilibrium & Ionic Equilibrium (Question Bank # 02)

1. **Ans. (B)**

Sol.

	CH_3COOH	+ $NaOH$	$\longrightarrow CH_3COONa$	+ H_2O
Millimoles in 100 ml buffer	2	0.5	1	
	1.5	x	1.5	

For acidic buffer

$$pH = pKa + \log\left(\frac{5}{a}\right)$$

$$= 5 - \log 2 + \log\left(\frac{1.5}{1.5}\right)$$

$$= 4.7$$

2. **Ans. (B)**

Sol.

	$PCl_5(g)$	$\rightleftharpoons PCl_3(g)$	$+ Cl_2(g)$
at t = 0;	1	–	–
At equilibrium	1 – α	α	α

Avg. M_W of mixture

$$\bar{M} = \frac{M}{1+\alpha}$$

$$\text{But } d = \frac{P\bar{M}}{RT}$$

$$d = \frac{PM}{RT(1+\alpha)} \Rightarrow 1+\alpha = \frac{PM}{dRT}$$

$$\alpha = \frac{PM}{dRT} - 1$$

3. **Ans. (B)**

Sol.
$$\begin{array}{lccc} & Na_2COONH_4(s) \rightleftharpoons & 2NH_3(g) & +\ CO_2(g) \\ \text{At I equilibrium:} & & 2P & P \\ \text{At new equilibrium:} & & 3P & P^1 \end{array}$$

as temperature is not charged so K_P is constant.

$$K_P = (2P)^2 \times P = (3P)^2 \times P^1$$

$$P' = \frac{4}{9}P$$

$$\frac{P_{total}(\text{new equilibrium})}{P_{total}(\text{original or I equilibrium})} = \frac{\left(3P + \frac{4}{9}P\right)}{2P + P} = \frac{\frac{31}{9}P}{3P}$$

4. **Ans. (C)**

Sol. (A)
$$\begin{array}{lcccc} & CH_3COOH & +\ NH_4OH & \longrightarrow CH_3COONH_4 & +\ H_2O \\ \text{m moles} & 3 & 3 & - & \\ & x & x & 3 & \end{array}$$

Salt of W.A & W.B. ⇒ Buffer

(B)
$$\begin{array}{lccc} & Na_2CO_3 & +\ H_2CO_3 & \longrightarrow 2NaHCO_3 \\ \text{m moles} & 2 & 4 & - \\ & x & 2 & 4 \end{array}$$

H_2CO_3 & $NaHCO_3$ from buffer.

(C)
$$\begin{array}{cccc} HCl & +\ Na_3PO_4 & \longrightarrow Na_2HPO_4 & +\ H_2O \\ 3 & 1 & - & \\ 2 & x & 1 & \end{array}$$

$$\begin{array}{cccc} Na_2HPO_4 & +\ HCl & \longrightarrow NaH_2PO_4 & +\ H_2O \\ 1 & 2 & - & \\ x & 1 & 1 & \end{array}$$

$$\begin{array}{cccc} NaH_2PO_4 & +\ HCl & \longrightarrow H_3PO_4 & +\ H_2O \\ 1 & 1 & - & \\ x & x & 1 & \end{array}$$

Only H_3PO_4 remain, so no buffer

(D)
$$\begin{array}{cccc} NH_4Cl & +\ NaOH & \longrightarrow NH_4OH & +\ NaCl \\ 1 & 0.5 & - & - \\ 0.5 & x & 0.5 & 0.5 \end{array}$$

NH_4OH & NH_4Cl form buffer.

5. Ans. (A)

Sol.

	$nA(g) \rightleftharpoons$	A_n
t = 0; moles	1	–
at equilibrium	$1-\alpha$	$\frac{\alpha}{n}$

$$\frac{PV}{RT} = n_{total} = 1-\alpha+\frac{\alpha}{n} = 1-\left(1-\frac{1}{n}\right)\alpha$$

$$= 1-(n-1)\frac{\alpha}{n}$$

$$k_c = \frac{\left(\frac{\alpha}{nV}\right)}{\left(\frac{1-\alpha}{V}\right)^n} = \frac{\alpha}{n}\ V^{n-1} \qquad \left(\text{given } \alpha \text{ is very small} \Rightarrow (1-\alpha) \simeq 1\right)$$

$$\frac{\alpha}{n} = \frac{k_c}{V^{n-1}}$$

6. Ans. (C)

Sol. (I) $\left[Ag^+\right] = \sqrt{k_{sp(AgCl)}} = 10^{-5}\ M$

(II) $\left[Ag^+\right] = \sqrt{k_{sp(AgI)}} = 10^{-8}\ M$

(III) $Ag^+(aq.) + 2NH_3(aq) \rightleftharpoons Ag(NH_3)_2^+(aq); k_f$

$$k_f = \frac{\left[Ag(NH_3)_2^+\right]}{\left[Ag^+\right][NH_3]^2}$$

$$10^8 = \frac{1}{\left[Ag^+\right](0.1)^2} \Rightarrow \left[Ag^+\right] = 10^{-10}\ M$$

(IV) $Ag^+(aq.) + 2CN^-(aq.) \rightleftharpoons Ag(CN)_2^-(aq.); k_f$

$$k_f = \frac{\left[Ag(CN)_2^-\right]}{\left[Ag^+\right]\left[CN^-\right]^2}$$

$$10^{11} = \frac{1}{\left[Ag^+\right](0.1)^2}$$

$$\left[Ag^+\right] = 10^{-13}\ M$$

7. **Ans. (D)**

Sol. $$PO_4^{3-} + H_2PO_4^- \longrightarrow 2HPO_4^{2-}$$

$$\begin{array}{ccc} 5 & 2.5 & - \\ 2.5 & x & 5 \end{array}$$

$$pH = pKa_3 + \log\frac{[PO_4^{3-}]}{[HPO_4^{2-}]}$$

$$= 12 + \log\left(\frac{2.5}{5}\right) = 12 - \log 2$$

8. **Ans. (B)**

Sol. $$Ag^+ + 2CN^- \rightleftharpoons Ag(CN)_2^-\ ;\ K_c = \frac{1}{4\times10^{-19}}$$

$$= \frac{1}{4}\times10^{19}$$

$$\begin{array}{llll} \text{At } t = 0; & 0.03 & 0.1 & - \\ \text{At equilibrium} & \sim & (0.1 - 0.06) & 0.03 \\ & & = 0.04 & \end{array}$$

$$\frac{1}{4}\times10^{19} = \frac{(0.03)}{[Ag^+](0.04)^2}$$

$$[Ag^+] = 7.5\times10^{-18}\text{ M}$$

9. **Ans. (A)**

Sol. Rate of hydrolysis; $r \,\alpha\, [H^+]$

$r = k[H^+]\,[\text{ester}]$

$$\frac{1}{100} = \frac{[H^+]_{HA}}{[H^+]_{HX}} = \frac{[H^+]_{HA}}{1}$$

$$[H^+] = 10^{-2}\text{M}$$

For weak acid, $k_{a(HA)} = \frac{[H^+]^2}{C-[H^+]} = \frac{(10^{-2})^2}{1-10^{-2}} = 10^{-4}$

10. **Ans. (C)**

Sol. $\Delta G^{o} = -RT\ell n\, k_p$

$-1743 = -8.3 \times 300\ \ell n\, k_p$

$\ell n\, k_p = 0.7$

$k_p = 2\ ;\qquad k_p = k_c(RT)^{\Delta ng}$

(A) $2 = k_c(0.082 \times 300)^{-1}$

$k_c = 2 \times (0.02 \times 300) = 49.2$

But from given data

$k_c = \frac{3}{3 \times 6} = \frac{1}{6}$

Both k_c values are not same

(B) $k_c = \frac{2}{(0.082 \times 300)^2} \neq| \frac{6 \times 3^2}{3}$

(C) $k_c = 2(R \times 300)^{o} = \frac{6 \times 3}{(3)^2}$

Both k_c data are same.

(D) $k_c = 2 \neq \frac{(3)^2}{3 \times 6}$

11. **Ans. (A,B,C)**

Sol. (A) W.B. + salt of W.B. & S.A. ⇒ buffer

(B) $$\begin{array}{cccc} CH_3COOH & + NaOH & \longrightarrow CH_3COONa & + H_2O \\ 2 & 1 & - & \\ 1 & x & 1 & \end{array}$$

W.A. + salf of W.A & S.B. ⇒ Buffer

(C) $$\begin{array}{ccc} NH_3 & + HCl & \longrightarrow NH_4Cl \\ 2 & 1 & - \\ 1 & x & 1 \end{array}$$

W.B. + salt of W.B. & S.A. ⇒ Buffer

(D) $$\begin{array}{cccc} CH_3COOH & + NaOH & \longrightarrow CH_3COONa & + H_2O \\ 1 & 2 & - & \\ x & 1 & 2 & \end{array}$$

⇒ not buffer

node06\

12. Ans. (B,D)

Sol. (A) $k_{sp} = \left[Fe^{+3}\right]\left[OH^-\right]^3 \Rightarrow \left[Fe^{+3}\right] \propto \frac{1}{\left[OH^-\right]^3}$

$\Rightarrow \left[H^+\right]\downarrow \Rightarrow \left[OH^-\right]\uparrow \Rightarrow \left[Fe^{+3}\right]\downarrow$

(B) $pOH = pK_b + \log\frac{[NH_4Cl]}{[NH_4OH]}$

$[NH_4Cl]\uparrow \Rightarrow pOH\uparrow \Rightarrow pH\downarrow$

(C) $NH_4I \rightarrow$ Acidic (salt of W.B. & S.A.)

$Cr(NO_3)_3 \rightarrow$ Acidic (salt of W.B. & S.A.)

$KCN \rightarrow$ Neutral (salt of S.A. & S.B.)

(D) Degree of hydrolysis for salts of W.A. & W.B. is independent of their concentration.

13. Ans. (B)

Sol. (A) $CH_3COOH \Rightarrow pH = \frac{1}{2}pka - \frac{1}{2}\log C$

on adding NaOH

$$\underbrace{CH_3COOH + \underset{(L.R.)}{NaOH} \longrightarrow CH_3COONa}_{Buffer} + H_2O$$

(B) $0.1\,M\ NaHCO_3 \longrightarrow pH = \frac{pka_1 + pka_2}{2}$

(C) NH_3 increase solubility of AgCl by complex formation.

(D) $NH_3 + HCl \longrightarrow NH_4Cl$

(L.R.)

✓ × ✓

form buffer

14. Ans. (C)

Sol. $CH_3COOH + NaOH \longrightarrow CH_3COONa + H_2O$

CH_3COOH	$NaOH$	CH_3COONa
3	1	–
2	×	1

$pH = pka + \log\frac{s}{a} = pka + \log\left(\frac{1}{2}\right)$

$= (5 - \log 2) - \log 2 = 4.4$

ALLEN®

15. Ans. (A)

Sol. $\left[H^+\right]=\sqrt{ka_1+Ka_2}=10^{-9};\left[OH^-\right]=\frac{10^{-14}}{10^{-9}}=10^{-5}\ mol/L$

INTEGER TYPE

1. Ans. (1.00)

Sol. $HCl+NaOH\longrightarrow NaCl+H_2O$ (First reaction occur between S.A. & S.B.)

m moles	2	1	–
	1	x	1

Final concentration after mixing of solutions.

$$[HA]=\frac{10\times0.1}{30}=\frac{1}{30}M$$

$$[HCl]_{left}=\frac{1}{30}M$$

$HA(aq.)$	$\rightleftharpoons$	$H^+(aq.)$	$+$	$A^-_{(aq)}$; $k_a=10^{-5}$
$\frac{1}{30}$		$\frac{1}{30}$		–
$\left(\frac{1}{30}-x\right)\simeq\frac{1}{30}$		$\left(\frac{1}{30}+x\right)\simeq\frac{1}{30}$		x

(as $\frac{Ka}{C}<10^{-3}\Rightarrow$ x is negligible compare to $\frac{1}{30}$)

$$K_a=\frac{[H^+][A^-]}{[HA^-]}=\frac{\frac{1}{30}\times[A^-]}{\frac{1}{30}}$$

$$\left[A^-\right]=K_a=10^{-5}\ M$$

$$=1\times10^{-5}\ M$$

$$\boxed{x=1}$$

2. **Ans. (3.20)**

Sol. $k_{sp} = 4s^3 = 4\times\left(2\times10^{-6}\right)^3 = 32\times10^{-18}$

In buffer solution. (solubility = x mol/L)

$$Pb(OH)_2(s) \rightleftharpoons Pb^{+2}(aq) + 2OH^-_{(aq)};\ k_{sp}$$

$$x \qquad \left(2x+10^{-6}\right)$$

$$\nearrow$$

due to buffer

$$\simeq 10^{-6}$$

$$K_{sp} = \left[Pb^{+2}\right]\left[OH^-\right]^2$$

$$32\times10^{-18} = x\left(10^{-6}\right)^2$$

$$x = 3.2\times10^{-5}\ M$$

Buffer does not allow $[OH^-]$ to change.

3. **Ans. (1.00)**

Sol. $P_{He(I)} = P_{He(II)} = 2\ atm$

I chamber

$$4 = \cancel{P_{NH_3}} + \cancel{P_{H_2S}} + \cancel{P_{He(I)}}$$

$$P \qquad P$$

$$\Rightarrow P = 1\ atm$$

$$NH_4HS(s) \rightleftharpoons NH_3(g) + H_2s(g);\ kp$$

$$P \qquad\qquad P$$

$$kp = 1\times1 = 1$$

4. **Ans. (4.00)**

Sol. At STP $\Rightarrow v \propto n$ (same P & T)

$$2COF_2(g) \rightleftharpoons CO_2(g) + CF_4(g)$$

Ratio of moles at equilibrium	100	200	200
Partial pressure ($P \propto n$)	2	4	4

$$V_{CF_4} = 500 - 300 = 200\ ml$$

$$\Rightarrow k_p = \frac{4\times4}{2^2} = 4$$

ALLEN®